

2. Initiating Project

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Q.1 What are non-numeric models of project selection? Explain in brief.

⇒

- Project selection is an important function of business organization.
- A wrong selection of project may generate losses which is undesirable.
- Non-numeric models are qualitative methods used to evaluate and prioritize projects without using financial or quantitative data.
- These models are generally subjective and rely on judgement, experience rather than measurable metrics.

Types of non-numeric models include

- 1) The Sacred Cow
- 2) Operating Necessity
- 3) Competitive Necessity
- 4) Product Line Extension
- 5) Comparative Benefit Model
- 6) Q-sort Model

1) Sacred Cow

⇒ This is a commonly used method in which the senior members of organization suggest a project out of their own understanding and expertise.

- Most of the projects using this method are those that are initiated out of an apparent opportunity available for new product.

- The project is created as an immediate result of this approach in which about whatever the boss proposes becomes the project.

- The sacredness of such project reflects that it generally continues till it finishes or until the senior declares it as a failure of the idea & ends it.

- For e.g., A CEO wants to launch a project to redesign the company's website.

2) Operating Necessity

⇒ Sometimes, the ~~efficiency~~ necessity of a project arises simply because it is capable of either enhancing the operation efficiency of the organization or solving a serious problem that's troubling the organization.

- For e.g., a plant is threatened by the flood then it is not much complex and effortful to start a project for developing a protective desk about flood.

- Many potential projects are evaluated for selection using this criterion of project

3) Competitive Necessity

⇒ Organization always like to stay ahead in competition and this desire becomes the basis for decision making decision to carry out a project.

- Although, preference is generally given to the project whose needs is felt by

the operating necessity to competitive necessity with regard to investment.

- For e.g., Developing a mobile app because competitors are gaining user through theirs.

4) Product Line Extension

- ⇒ Projects are judged on how they fit with current product line, fill a gap, strengthen a weak line or extend the line in a new desirable way.
- Ensures brand consistency and caters to evolving customer demands.
- For e.g., Launching a new Flavor of popular snack product.

5) Comparative Benefit Model

- ⇒ This model is used when there are several projects up of consideration by the organization.
- The selection is done on the basis of their comparative studies and only that project is selected which can provide the maximum benefit to the model.
- Though, comparing various projects is not an easy task as there are chances of committing mistakes.
- For e.g., categorizing projects into high/medium/low priority based on stakeholder input.

6). Q-sort Model

⇒ This is one of the most simple and straightforward techniques used for selecting projects.

- In this model, the available projects for undertaking are first divided into three groups according to their relative merits. They are: Good, Fair and Poor.

Q.2 Explain different numeric models of project selection.

⇒

- Models that return numeric value for a project that can be easily compared with other projects.

1) Payback period

⇒ This method is a discount simple non-discounted factor method.

- It calculates the time period that will be required for the project to generate enough returns to recover the initial investment.

Formula :-

$$\text{Payback period} = \frac{\text{Project cost}}{\text{Annual cash flow}}$$

For e.g.,

A project costing \$100,000 with an annual cash inflow of \$25,000 has a payback period of 4 years.

2) Net present value (NPV) Model

⇒ NPV is calculated by subtracting the present value of cash outflows from the present value of cash inflows.

The present value defines what will be the worth of the future sum of money as of today.

Formula :-

$$NPV = A_0 + \sum_{t=1}^N \frac{F_t}{(1+k)^t}$$

Here,

A_0 = Initial cash investment,

F_t = cash flow in time period t

r = The discount rate

t = no. of years of life

For e.g.,

A project with an NPV of \$50,000 is more attractive than one with an NPV of \$10,000.

3) Internal Rate of Return (IRR) Model

⇒ The Internal rate of return is a discounting cash flow technique.

It is the discounting rate at which the NPV is equal to zero.

Formula :-

$$IRR = \frac{\text{Cash Flow}}{(1+r)^t} - \text{Initial investment}$$

Here,

r = discount rate

t = time period

4) Profitability Index

⇒ Also known as the benefit-cost ratio, the profitability index is the net present value of all future expected cash flows divided by the initial cash investment.

If this ratio is greater than 1.0, the project may be accepted.

Q.3 Explain the significance of IRR method in project selection.

⇒

- The internal rate of return is a discounting cash flow technique.
- It gives a rate of return as expected to be earned by a project.
- It gives a discounted rate where the total of initial cash outlay and discounted cash inflows are equal to zero.
- In other words, it is discounting rate at which the net present value (NPV) is equal to zero.
- Formula:

$$IRR = \left(\frac{\text{Cash Flows}}{\text{Initial Investment}} \right)^{1/i}$$

where,

r = discount rate

i = time period.

• Significance :

1) Measures Profitability

- ⇒ IRR directly indicates the project's earning potential.
- A higher IRR implies a more attractive project.

2) Aids Decision-making

- ⇒ IF $IRR > \text{cost of capital}$ the project is financially viable.
- Helps compare multiple project efficiently

3) Time value of Money considered

- ⇒ Unlike simple ROI, IRR considers the time value of money, which makes it more accurate.

4) Helps in ranking project

- ⇒ Projects can be ranked based on IRR when capital is limited.
- Useful in capital rationing.

5) Useful in long-term investment

- ⇒ Especially useful in evaluating Infrastructure, IT systems or manufacturing expansion projects.

Q.4 Consider a project having following cash flow stream. The cost of capital (r) for the firm is 20%. Calculate NPV of project and decide whether to accept or reject the project.

Year	0	1	2	3	4	5
Cash Flow	10,00,000	2,00,000	2,00,000	3,00,000	3,00,000	3,50,000

⇒

Soln:-

$$r = 20\% = 0.2$$

— Present value Formula:-

$$PV = \frac{CF_0}{(1+r)^0} + \frac{CF_1}{(1+r)^1} + \frac{CF_2}{(1+r)^2} + \frac{CF_3}{(1+r)^3} + \frac{CF_4}{(1+r)^4} + \frac{CF_5}{(1+r)^5}$$

$$PV = \frac{\text{Cash Flow}}{(1+r)^t}$$

$$= \frac{10,00,000}{(1+0.2)^0} + \frac{2,00,000}{(1+0.2)^1} + \frac{2,00,000}{(1+0.2)^2} + \frac{3,00,000}{(1+0.2)^3} + \frac{3,00,000}{(1+0.2)^4} + \frac{3,50,000}{(1+0.2)^5}$$

$$= 10,00,000 + 1,66,666.67 + 1,38,888.89 + 1,65,289.26 + 1,37,741.51 + 1,83,823.53$$

— Calculate PV:-

$$PV_0 = \frac{10,00,000}{(1+0.2)^0}$$

$$i) PV_0 = \frac{10,00,000}{(1+0.2)^0} = 10,00,000$$

$$\therefore PV_0 = 10,00,000$$

Calculate PV of cash flows at year 1, 2, 3, 4, 5

$$ii) PV_1 = \frac{2,00,000}{(1+0.2)^1} = 1,66,666.67$$

$$iii) PV_2 = \frac{2,00,000}{(1+0.2)^2} = 1,38,888.89$$

$$iv) PV_3 = \frac{3,00,000}{(1+0.2)^3}$$

$$= 225394.4403$$

$$v) PV_4 = \frac{3,00,000}{(1+0.1)^4}$$

$$= 2004,904.0366$$

$$vi) PV_5 = \frac{3,50,000}{(1+0.1)^5}$$

$$= 2,17,322.4631$$

$$\therefore NPV = PV_0 + PV_1 + PV_2 + PV_3 + PV_4 +$$

$$PV_5$$

$$= -10,00,000 + 181818.1818 +$$

$$(65289.2562 + 225394.4403$$

$$+ 2004904.0366$$

$$+ 217,322.4631$$

$$\therefore NPV = -5271.6220$$

$\therefore$ NPV value is negative.

Hence, the project will be rejected.

Q.5 (Dec 23) Q.2 a

⇒ Solⁿ:-

Cash Flow	Project A	Project B	Project C	Project D
0	-20,00,000	-20,00,000	-20,00,000	-20,00,000
1	5,00,000	6,00,000	10,00,000	3,00,000
2	5,00,000	6,00,000	8,00,000	5,00,000
3	5,00,000	6,00,000	6,00,000	7,00,000
4	5,00,000	6,00,000	4,00,000	9,00,000
5	5,00,000	6,00,000	2,00,000	11,00,000
Discount rate	6%	9%	13%	12%

∴ PV = Cash Flow

$$= \frac{CF_t}{(1+r)^t}$$

• Project A :

$$r = \frac{6}{100} = 0.06$$

$$PV_0 = \frac{-20,00,000}{(1+0.06)^0} = -20,00,000$$

$$PV_1 = \frac{5,00,000}{(1+0.06)^1} = 471,698.1132$$

$$PV_2 = \frac{5,00,000}{(1+0.06)^2} = 444,998.2200$$

$$PV_3 = \frac{5,000,000}{(1+0.06)^3} = 419,809.6415$$

$$PV_4 = \frac{5,000,000}{(1+0.06)^4} = 396,046.8316$$

$$PV_5 = \frac{5,000,000}{(1+0.06)^5} = 373,629.0864$$

$$\begin{aligned} NPV_A &= -20,000,000 + 471,698.1132 + 444,998.22 \\ &\quad + 419,809.6415 + 396,046.8316 \\ &\quad + 373,629.0864 \\ &= 106,181.8927 \end{aligned}$$

Project B

$$r = \frac{9}{100} = 0.09$$

$$PV_0 = \frac{-20,000,000}{(1+0.09)^0} = -20,000,000$$

$$PV_1 = \frac{6,000,000}{(1+0.09)^1} = 5,504,58.7156$$

$$PV_2 = 5,050,07.8960$$

$$PV_3 = 4,633,10.0880$$

$$PV_4 = 425055.1266$$

$$PV_5 = 389358.8318$$

$$\therefore NPV_B = 333790.7580$$

• Project C

$$r = \frac{15}{100} = 0.15$$

$$PV_0 = \frac{-20,00,00,000}{(1+0.15)^0} = -20,00,000$$

$$PV_1 = \frac{10,00,000}{(1+0.15)^1} = 869,565.2714$$

$$PV_2 = \frac{800,000}{(1+0.15)^2} = 604,914.9338$$

$$PV_3 = \frac{6,00,000}{(1+0.15)^3} = 394,509.7335$$

$$PV_4 = \frac{4,00,000}{(1+0.15)^4} = 228,701.2982$$

$$PV_5 = \frac{2,00,000}{(1+0.15)^5} = 99,435.3471$$

$$\therefore NPV_C = 197,126.5900$$

• Project D $r = 10\%$

$$PV_0 = -20,00,000$$

$$PV_1 = 245901.6393$$

$$PV_2 = 335,931.2013$$

$$PV_3 = 385,494.8212$$

$$PV_4 = 406,259.1792$$

$$PV_5 = 406,999.1777$$

$$\therefore NPV_A = -219,413.19813$$

$$\therefore NPV_B > NPV_C > NPV_A > NPV_D$$

$$NPV_B > NPV_C > NPV_A > NPV_D$$

$\therefore$ Project B will be selected.

(Q. 9 b) (May 24)

Solⁿ:

Year	Cash Flow	
0	-200000	$\therefore PV = \frac{\text{Cash Flow}}{(1+r)^t}$
1	10000	
2	10000	
3	10000	$\therefore r = 12\% = 0.12$

$$\therefore PV_0 = \frac{-200000}{(1+0.12)^0} = -200000$$

$$\therefore PV_1 = \frac{10000}{(1+0.12)^1} = 8928.5714$$

$$\therefore PV_2 = \frac{10000}{(1+0.12)^2} = 7971.9387$$

$$\therefore PV_3 = \frac{10000}{(1+0.12)^3} = 717.9443$$

$$\therefore NPV = PV_0 + PV_1 + PV_2 + PV_3$$

$$= -200000 + 8928.5714 + 7971.9387 + 717.9443$$

$$\therefore NPV = -1,75,981.5456$$

→ Not accepted

Q.6 Explain process of project portfolio and project charter.

⇒

- The Project portfolio process is a method which is used to maximize the output potential of all projects undertaken by an organization at a given time.
- The PPP attempts to link the organization's projects directly to the goals and strategy of the organization.
- PPP is also means for monitoring and controlling the organization's strategic projects.
- Steps

1) Establish a project council

⇒ The main purpose of the project council is to establish and articulate a strategic direction for those projects spanning internal or external boundaries of the organization.

- Thus, senior managers must play a major role in this council.

- Without the commitment of senior management, the PPP will be incapable of achieving its main objective.

2) Identify project categories & criteria

⇒ In this step, project categories are identified so the mix of projects funded by organization will be spread across appropriately across these areas making major contribution to goals.

- In addition, within each category, criteria are established to discriminate between very good and even better projects.

- The criteria are also weighted to reflect their relative importance.

3) Collect the project data

⇒ For existing and proposed project, assemble the data appropriate to that category's criteria.

- Be sure to update data for ongoing projects, and not just use the data from previous evaluation.

- For cost data, use "activity based costs" rather than incremental costs.

4) Assess resource availability

⇒ Assess the availability of both internal and external resources, by type, timing and department.

- It detects overcommitments or limitations in capacity.

- Match the resources with project demands to avoid bottlenecks.

5) Reduce the project and criteria set

⇒ In this step, multiple screens are employed to narrow down the number of competing projects.

- The result of this step may involve cancelling some ongoing projects or replacing them with new, more promising projects.

- Simplifies evaluation and selection process.

6) Prioritize the projects within categories
 ⇒ Apply the scores and criterion weights to rank the projects within each category.

- When checking the result of this step, however, reconsider the projects in terms of their benefits first and their resource costs second.

7) Select project to be funded & held in reserve

⇒ Choose the top-category priority projects for immediate implementation.

- Keep others in a standby/reserve list for future consideration.

- Ensures flexibility and readiness for changing circumstances.

8) Implement the process

⇒ Launch selected projects and assigns resources accordingly.

- Monitor progress using tools like KPIs, reports or dashboards.

- Continuously improve the process based on performance feedback.

Project Charter

- Project charter is a statement that describes the objective of a project.
- This statement also sets out detailed projects goals, roles and responsibilities, identifies main stakeholder and the level of authority of project manager.
- It also acts as a guideline for future projects.

Contents of Project Charter

- 1) Project purpose or justification
- 2) High-level requirements
- 3) Assumptions and constraints
- 4) Stakeholder list
- 5) Summary budget
- 6) Measurable project objectives and related success criteria.
- 7) Project benefits
- 8) Risk and issues
- 9) Project manager and authority level
- 10) Milestone schedule

Steps

1) Initiate the project

→ Gather input from sponsors and stakeholders.

- Identify strategic need.

2) Develop the Charter Document

- ⇒ Define scope, objective, stakeholders and roles.
- Document constraints, assumptions and key risks.

3) Review and Approval

- ⇒ Present document to sponsor for validation.
- Get formal sign-off.

4) Distribute the Charter

- ⇒ Share with team and stakeholders to align understanding and expectations.
- Typically, project manager prepares the charter, often with input from stakeholder, sponsors and senior management.
- The Project sponsor or Senior Executive authorizes and signs the charter, officially approving the project.

Q.7 Short note on : Project Proposal

⇒

- The set of documents submitted for evaluation is called the project proposal, whether it is brief or extensive, and regardless of the formality with which it is prepared/presented.
- It helps organizations and external project proposer stakeholders agree on an initial project planning framework.
- There are two types of project proposal
 - 1) Solicited
 - 2) Unsolicited

- Solicited project proposals are requested by prospective client.
- While unsolicited project proposals, you approach a potential sponsor with a proposal, even if they don't request one, to gain their business.

• Project Proposal Tips

1) Keep it simple

- ⇒ While there's no ideal project proposal length, focus on quality over quantity.
- Keep sentences short and simple, and avoid the use of business jargon.

2) Stay on-brand

⇒ Don't be afraid to let your company's personality shine through in your proposal.

Stay true to your brand and show the sponsor what sets you apart from your competitors.

3) Include data and visuals

⇒ Don't forget to include compelling, quantitative data.

When applicable, use visuals such as charts and graphs to enhance the proposal.

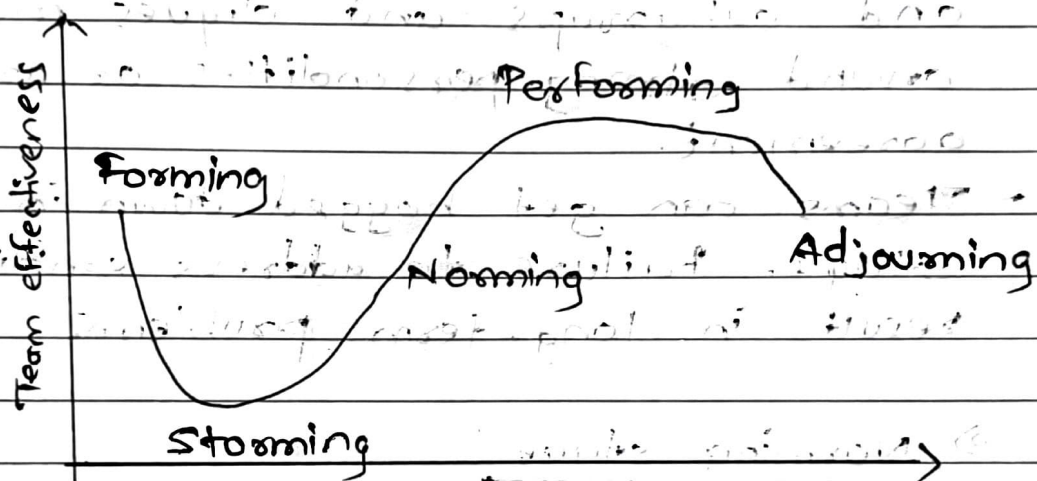
4) Quality control

⇒ Before you send the proposal out, make sure to read it and re-read it for any typos or grammatical errors.

Q.8 What are four stages of team development and growth? What are the barriers to team effectiveness?

⇒

- This process of learning to work together effectively is known as team development.
- Research has shown that teams go through definitive stages during team development.



1) Forming stage

- ⇒ The forming stage involves a period of orientation and acquaintance.
- There is a high level of uncertainty during this stage and people look for leadership and authority.
 - Most interactions are social as members get to know each other.
 - In this stage, team members ask for their roles and benefits.
 - A member who asserts authority or is knowledgeable may be looked to take control.

2) Storming stage

⇒ The storming stage is the most difficult and critical stage to pass through.

- It is a period marked by conflict and competition as individual personalities emerge.
- Team performance may actually decrease in this stage because energy is put into unproductive activities.
- Members may disagree on team goals, and subgroups and cliques may form around strong personalities or areas of agreement.
- Teams can get bogged down in this stage. Failure to address conflicts may result in long-term problems.

3) Norming stage

⇒ By getting through the norming stage, conflict is resolved and some degree of unity emerges.

- In the norming stage, consensus develops around who the leader or leaders are and individual members' roles.
- Interpersonal differences begin to resolve and a sense of cohesion and unity emerges.
- Team performance increases during this stage as members learn to cooperate and begin to focus on team goals.

4) Performing Stage

- ⇒ In this stage, consensus and cooperation are well-established and team is mature, organized and well-functioning.
- There is a clear and stable structure, and members are committed to the team's mission.
 - Problems and conflicts will emerge, but they are dealt with constructively.
 - The team is focused on problem solving and meeting team goals.

5) Adjourning Stage

- ⇒ In the adjourning stage, most of the team's goals have been accomplished.
- The emphasis is on wrapping up final tasks and documenting the effort and result.
 - As the work load is diminished, individual members may be reassigned to other teams, and the team is disbanded.
 - There may be regret as the team ends, so a ceremonial acknowledgement of the work and success of the team can be helpful.

• Barriers to team effectiveness

1) Poor communication

⇒ Leads to misunderstandings, delays and frustration.

2) Lack of trust

⇒ Team members may withhold ideas or avoid collaboration.

3) Ineffective leadership

⇒ Weak guidance can result in poor performance or conflicts.

4) Resistance to change

⇒ Team members may be ~~and~~ reluctant to adapt to new methods.

Q.9 Note on : Project Sponsor.

⇒

- A project sponsor is a person or group that provides resources and support for the project, programs or portfolio and is accountable for enabling its success.
- The project sponsor is a very important part of the project management chart.
- The project sponsor is one level above the project manager.
- They do not manage the day to day operations of the project but ensures the resources are in place, promote the project and hold overall responsibilities for the project's success.
- Roles and responsibilities in

1) Authorization

⇒ They authorize the project and assign work to the project manager.

2) Funding

⇒ They are responsible for ensuring that funding is in place and approve changes to the project budget.

3) Approving

⇒ They approve the project management plan and are kept aware of how the project is managed.

4) Promotion

⇒ They attempt to keep the project at the highest priority within the organization by promoting them.

5) Project charter

⇒ The document officially creates the project and assigns the project manager.

Q.10 Explain the aggregate project plan used in the portfolio process.

⇒

• An aggregate project plan is a strategic tool used to organize, prioritize and balance all projects within a company's portfolio.

• It ensures that the mix of projects aligns with business goals, optimizes resource utilization and manages risk across the portfolio.

• Purpose :

1) Align project with strategy

⇒ Ensures every project supports the organization's strategic goals.

2) Balance Project types

⇒ Includes a mix of high-risk, innovative, low-risk and improvement projects.

3) Manage Capacity and Resources

⇒ Matches project load with available manpower, budget and time.

4) Enhance decision-making

→ Helps top management decide which projects to invest in or terminate.

Steps in creating an Aggregated Project Plan:

1) Identify all current and proposed projects

⇒ Gather information on all existing and suggested projects within the organization.

- Consider projects from R&D, marketing, IT, operations, etc.

- Remove duplicates and non-feasible ideas

from the list before proceeding.

2) Categorize Projects into types

⇒ Sort the projects into types such as

Breakthrough, Platform, Derivative,

R&D and support, etc.

- Understand the goal of each project to fit it into the right category.

- Determine how each project contributes to innovation, improvement or support.

3) Assess strategic fit & Resource Requirement

⇒ Ensures each project aligns with organizational goals and mission.

- Calculate manpower, time and budget needs for each project.

- Look for resource overlaps or dependencies across projects.

4) Prioritize projects based on criteria

⇒ Use factors like ROI, risk, market impact, time-to-completion, etc.

- Apply weighted scores or ranking techniques to assess project value.
- Involve decision-makers to ensure balanced prioritisation.

5) Balance portfolio by type, risk and return

⇒ Ensures a balanced mix of short-term, long-term, high-risk and safe projects.

- Prevent overcommitting resources to one type of project or department.
- Combine breakthrough innovation with stable, low-risk projects.

6) Regularly review and update the plan

⇒ Track ongoing project performance against goals and timelines.

- Revise the portfolio when strategic priorities or market condition shifts.
- Incorporate learning and feedbacks into future project planning.

Q 19 List and explain the criteria for choice of project selection model.

⇒

- Selecting the right project is crucial for organizational success.
- A project selection model helps evaluate and prioritize projects based on strategic goals, feasibility and expected benefits.
- The key criteria include:

1) Alignment with Organizational Goals

⇒ The project must support the company's strategic objectives.

- For e.g., A tech company prioritizing AI development will select projects related to AI innovations.

2) Profitability & Return on Investment

⇒ The model should evaluate financial benefits, such as Net Present Value (NPV), Internal Rate of Return (IRR) or Payback period.

- For e.g., A business will prioritize a project with a high ROI over one with uncertain returns.

3) Resource Availability

⇒ Ensure the project can be executed with available humans, financial and technical resources.

- For e.g., A company with limited skilled manpower will avoid highly labor-intensive projects.

4) Risk Assessment

⇒ The model should consider potential risks such as financial, technical, operational, etc.

- For e.g., A construction firm may avoid projects in politically unstable regions due to high risk.

5) Scalability

⇒ The model should assess whether the project can expand and deliver long-term value.

- For e.g., A telecom company investing in 5G infrastructure for future network expansion.

Q.12 What is Team Dynamics?

⇒

- Team dynamics is a broad concept and represents the way in which team members behave.
- These dynamics impact communication, collaboration, motivation and overall team success.
- Key elements of team dynamics :-

1) Communication

⇒ Effective communication is crucial for successful team dynamics.

- This includes listening actively, expressing ideas clearly and providing feedback.

2) Trust

⇒ Team members must feel comfortable and safe while sharing ideas and information without fear or embarrassment.

- Established trust leads to better ideas and creativity.

3) Roles and responsibilities

⇒ Clearly defined roles ensure accountability and efficiency.

- For e.g., A project manager oversees planning, while developers focus on execution.

4) Effective leadership

⇒ When you are able to check off all the necessary items, you are doing your part in building an effective team dynamic.

- As leader, you should do the following:

- i) Set clear goals for each team member and the team.
- ii) Share examples and experiences of your personal successes and mistakes in order to relate to the team.

5) Goal clarity

⇒ Organizational goal clarity keeps everyone moving in the same direction.

- Goal clarity facilitates a number of valuable outcomes:-

- i) It allows your people to make more of their own decisions.
- ii) Fewer issues will need to be escalated to you to resolve.
- iii) Decisions will be made more quickly.